
Mid-LN: Stabilizing Transformer Training via Layer Normalization Placement

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Abstract

The remarkable success of Transformer models across diverse domains is primarily driven by their core architecture, centered on attention mechanisms and layer normalization (LN). In this study, we propose the Mid-LN architecture, a novel building block that positions the LN layer immediately prior to the residual addition. This design is informed by an extensive investigation into how LN placement influences training stability and overall performance. By normalizing sublayer outputs before they are integrated into the residual pathway, the Mid-LN approach effectively stabilizes output distributions and suppresses the amplification of variance during backpropagation. Additionally, our attention scaling experiments reveal that learnable scaling yields the most stable gradient characteristics and superior convergence compared to other scale transformations. Empirical results demonstrate that Mid-LN significantly outperforms the standard Pre-LN architecture across key stability and performance metrics.

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Keywords: Layer Normalization, Attention Scaling, Architecture, Transformer.

1 Introduction

Since its introduction [1], the Transformer model has emerged as a predominant force in various realms of deep learning. In particular, it has demonstrated performance in natural language processing [2, 3], image analysis [4, 5], and various other multimedia applications [6, 7, 8], establishing itself as the standard for modern deep learning model architectures. Based on its easily parallelizable structure, powerful representational capacity through attention mechanisms, and scalability, the Transformer has evolved into large scale models such as PaLM, LLaMA, and Falcon, reaching the capability to train on trillions of tokens. Meanwhile, the Transformer architecture has been employed in various downstream tasks, including named entity recognition (NER) [9, 10], neural machine translation (NMT) [11, 12, 13], sentiment classification [14], and general language model construction [15], owing to its broad applicability and powerful representational capacity.

Despite being widely utilized and popular, the mechanisms through which the individual components of a Transformer transform input into output remain somewhat opaque. Numerous studies are dedicated to unraveling the intricacies of in block elements such as Feed-Forward Networks (FFN), Attention Mechanisms, Residual Connections, and Layer Normalizations (LN), seeking to understand their collective impact on overall performance. One specific focus is the arrangement of the LN within a Transformer block. The incorporation of LN is crucial to the architecture, playing a pivotal role in stabilizing the training process and augmenting the model’s overall performance for larger datasets. This stabilization, achieved by normalizing layer activations, proves essential due to significant variations in gradients across each block, presenting challenges in the optimization process.

However, most existing research has focused on training efficiency techniques such as a data sorting algorithm [16] or limited direct attempts to improve NMT performance through structural modifications of the architecture itself including Pre-Layer Normalization (Pre-LN). Accordingly, this study designed a novel Mid-LN structure. This structure is expected to improve distributional stability by normalizing the outputs of lower layers, thereby maintaining subsequent residual flows in a more consistent form.

The experiment results demonstrate that the proposed Mid-LN structure significantly improves training stability compared to Pre-LN, suppresses gradient explosion, and achieves more stable convergence. By exhibiting superior translation performance across various experiments, it was confirmed that this structure serves as a practical alternative that simultaneously

enhances both the stability and performance of Transformer training. Therefore, the structural improvements and experimental analyses proposed in this study are expected to not only address the limitations of existing Transformers but also directly contribute to improving the training stability and final performance of NMT models.

2 Background

2.1 Existing Models

In the initial structure of the Transformer, the Post-LN structure proposed by Vaswani et al. [1] was prevalent. Subsequently, research [17] introduced the Pre-LN structure, in which LN is applied to the values input to each sublayer. This was reported to improve the overall training stability and performance of the model.

However, subsequent theoretical studies [18, 19] continuously reported that as the depth of the Transformer increases, training instability persists even when the Pre-LN structure is applied [20]. As an alternative to address this issue, the B2T Residual structure was proposed, which propagates the output of the first layer to the final layer. However, this approach did not completely prevent gradient vanishing [21].

2.2 Layer Normalization

Layer normalization (LN) was introduced by Lei Ba et al. (2016) [22]. The paper addresses the problem of internal covariance shift. Layer normalization alleviated this problem by normalizing the activations of each layer before they are passed to the next layer. LN is defined as

$$\text{LN}(h) = \gamma \frac{h - \mu}{\sigma} + \beta \quad (1)$$

where h denotes the input vector, and γ and β represent the learnable scaling and bias parameters, $\mu = \frac{1}{d} \sum_{k=1}^d v_k$, and $\sigma^2 = \frac{1}{d} \sum_{k=1}^d (v_k - \mu)^2$. LN is known to improve training efficiency through stabilization of output distributions. However, it has also been noted that there exists a risk of overfitting if γ and β are not properly learned [23].

2.3 Scaled Dot-Product Attention

The scaled dot-product attention, a component of the Transformer, has been reported to exhibit the problem that the dot product of query and key vectors becomes large as the dimension d_k increases. Vaswani et al. (2017) [1] explained that when these values become large, the softmax input is located in a high variance region, resulting in an output distribution that is very sharp and overreacts to even minor changes.

To counteract this issue, Vaswani et al. introduced Attention Scaling, which divides the dot product values by $\sqrt{d_k}$, thereby demonstrating that the variance of the softmax input can be controlled and gradient explosion and distribution distortion can be prevented. This scaling controls the overall optimization stability of the Transformer architecture in conjunction with LN.

3 Proposed Methods

The original Transformer architecture [1] includes an encoder-decoder architecture. Both encoder and decoder consist of a stack of N blocks, which combine fully connected feed-forward (FF) layers, a multi-head attention layer, a normalization layer, and residual connections. Structures are categorized as Post-LN and Pre-LN according to the position of LN within the block [19]. This study defines a novel Mid-LN structure and aims to compare it with Pre-LN.

3.1 Pre-LN

Pre-LN, shown in Figure 1 (a), applies LN before entering the sublayer and adds the residual to the sublayer output.

$$\text{PreLN}(x) = x + F(\text{LN}(x)) \quad (2)$$

This structure represented in (2), performs normalization at the input stage ensuring stability during the early phases of training [19]. However, the final output of the sublayer remains unnormalized, and during backpropagation gradients with respect to that output accumulate along the residual path. This creates the possibility of gradient explosion as the network deepens. Additionally, in later stages of training, there are phenomena of increased loss fluctuation or unstable convergence [18, 19, 20].

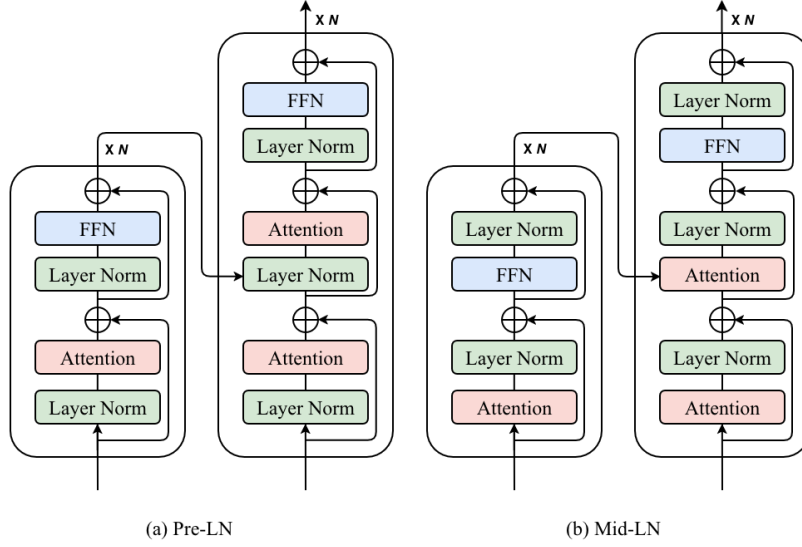


Figure 1 The existing Pre-LN and the newly proposed Mid-LN structure

3.2 Mid-LN

To mitigate these issues, this study proposed the Mid-LN structure, which applies LN to the output stage of sublayers. As shown in Figure 1 (b), Mid-LN applies LN after the Attention or FFN operation and combines its output with the residual path. This approach stabilizes the distribution of sublayer outputs, which reduces fluctuations during the training process and induces faster model convergence. Specifically, although the residual term itself is not normalized, the overall variance of gradients remains stable because the added term is always in normalized form.

$$\text{MidLN}(x) = x + \text{LN}(F(x)) \quad (3)$$

This structure in (3) does not completely eliminate gradient explosion. However, it suppresses abrupt expansion of gradient magnitudes between layers, which secures a more stable training curve compared to Pre-LN. Furthermore, unlike Post-LN, it does not normalize the entire output [23]. This allows the structure to maintain initial convergence speed while simultaneously securing stability.

3.3 Mid-LN Attention Scaling Design

In self attention, it is standard practice to scale the query and key dot product $QK^\top$ by $\sqrt{d_k}$ to stabilize the softmax input distribution [1]. This study treats this scaling coefficient as a design parameter and defines the following four variants for comparison.

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{s}\right) V, \quad s = \begin{cases} \sqrt{d_k}, \\ d_k, \\ \theta\sqrt{d_k}, \\ \theta d_k \end{cases} \quad (4)$$

In (4), $\theta > 0$ is a learnable scalar parameter that is shared across each self attention module and is initialized to 1. No separate normalization or range constraints are imposed on θ , and it is automatically updated by the loss function during backpropagation. In this study, scale correction was applied only to Attention and was not applied to other modules such as FFN.

4 Experiments

4.1 Model and Data

All experiments in this study were configured identically to the environment used in prior research. This study adopts the baseline experimental setup of the *Attention Is All You Need* paper. we use a Transformer model with the Pre-LN architecture and modify only the Mid-LN structure and the scaling factor s described Sections 3.2 and 3.3. Experiments were conducted over two weeks on a single NVIDIA GeForce GTX 1080 GPU [1]. For training, the maximum learning rate of the Adam optimizer was set to 5×10^{-4} , referencing the configuration proposed in [23], and the label smoothing coefficient was set to 0.01.

4.2 Comparison of Pre-LN and Mid-LN Structures

This study compared the stability of Pre-LN and Mid-LN by modifying the structure under identical training configurations. As shown in Figure 2 (a), in the case of Pre-LN, the gradient norm increased over the long term even after the initial rise, and intermittent spikes were also observed. In contrast, Mid-LN exhibited a pattern in which the gradient norm converged to a constant value as training progressed. This suggests that the Mid-LN structure

normalizes outputs at module boundaries, which suppresses scale amplification during backpropagation. Consequently, Mid-LN was confirmed to be a stable structure that effectively controls the effects of gradient accumulation, reduces loss fluctuations in the later stages of training, and maintains a more consistent gradient flow.

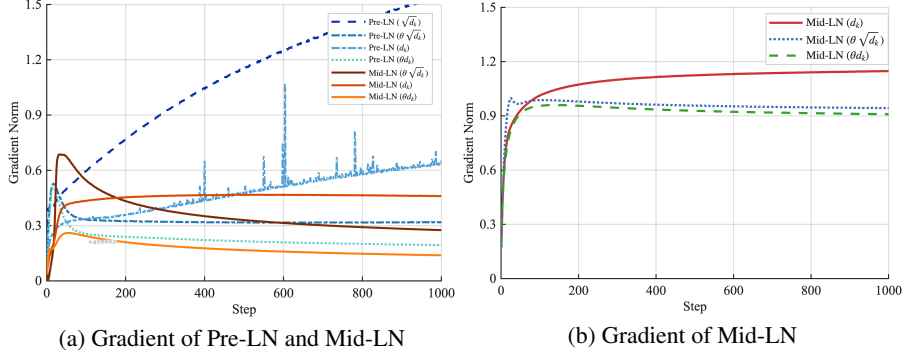


Figure 2 Gradient norm trajectories across training steps under different attention-scaling coefficients. (a) Encoder gradients for all compared variants. (b) Decoder gradients for Mid-LN variants only.

4.3 Comparison of Attention Scaling in Mid-LN

In this section, the structure was fixed to Mid-LN, and the scales $s \in \{d_k, \theta\sqrt{d_k}, \theta d_k\}$ defined in Section 3.3 were applied to the Mid-LN structure to comprehensively compare optimization stability and translation performance. The case of $\sqrt{d_k}$ was excluded from this comparison due to training instability. As shown by the red line in Figure 2 (b), in terms of training process stability, d_k exhibits instability in the later stages. It maintained a relatively higher gradient norm level compared to other models and showed a tendency toward reduced convergence efficiency. This can be interpreted as a scale mismatch that occurs because the fixed scale s fails to adapt despite the variance of $QK^\top$ fluctuating as training progresses. This mismatch deteriorates the conditioning of the softmax input logits and leads to softmax saturation and gradient instability.

4.4 Evaluations

We verified whether the numerical stability of the Mid-LN structure translates into actual translation performance improvement. The comparison targets

were set as the baseline Pre-LN($\sqrt{d_k}$) and Mid-LN(θd_k), which demonstrated the best stability in the analysis of Section 4.3.

Table 1 shows the bidirectional translation performance for Korean-English, the primary dataset of this study. Mid-LN exhibited a faster increase in BLEU during the early stages of training compared to Pre-LN, followed by a stable upward trend. To validate the generalization performance of the proposed structure, additional experiments were conducted using a German-to-English (DeEn) dataset. The results showed that Mid-LN (θd_k) achieved consistent performance improvements over the Pre-LN baseline across all translation directions, regardless of data scale and language pair.

Table 1 Translation performance comparison between Pre-LN and Mid-LN Transformers.

Model	En-Kr		Kr-En		De-En		En-De	
	BLEU	Loss	BLEU	Loss	BLEU	Loss	BLEU	Loss
Pre-LN	20.90 (19.83)	31.85	39.23 (36.55)	32.88	23.83 (24.40)	23.26	16.73 (19.34)	24.97
Mid-LN	21.48 (20.72)	29.95	39.72 (37.93)	31.23	24.27 (24.46)	22.33	16.34 (18.51)	24.78

5 Conclusions

This study analyzed the impact of LN position and attention scaling methods on training stability in Transformers. In the Pre-LN structure, sub-layer outputs were not normalized, resulting in gradient magnitudes that easily became unstable, which led to convergence instability in the later stages of training. In contrast, the Mid-LN structure normalized the distribution at the sublayer output, thereby maintaining stable gradient variance. As a result, the model converged more rapidly and more stably. Additionally, learnable θ based scaling automatically calibrated softmax inputs, contributing to improved numerical stability and translation performance. This study confirmed that the position of the LN is a critical design factor in the optimization of the transformer. In the future, we plan to explore the stability of deeper Transformer architectures by investigating various LN placement strategies.

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